

We see

$$\begin{aligned}
eQ_0(\delta_\chi\omega) &= -\frac{1}{3!}Q_0(\bar{\chi}\gamma^3 d_\omega\psi) - (L_\xi^\omega e - [c, e])\delta_\chi\omega + e\delta_\chi^2\omega \\
&= -\frac{1}{3!}(L_\xi^\omega \bar{\chi} - [c, \bar{\chi}])\gamma^3 d_\omega\psi + \frac{1}{3!}\bar{\chi}\gamma^3[\iota_\xi F_\omega - d_\omega c, \psi] \\
&\quad - \frac{1}{3!}\bar{\chi}\gamma^3 d_\omega(L_\xi^\omega \psi - [c, \psi]) - (L_\xi^\omega e - [c, e])\delta_\chi\omega + e\delta_\chi^2\omega \\
&= -\frac{1}{3!}L_\xi^\omega(\bar{\chi}\gamma^3 d_\omega\psi) + \frac{1}{3!}[c, \bar{\chi}\gamma^3 d_\omega\psi] - (L_\xi^\omega e - [c, e])\delta_\chi\omega + e\delta_\chi^2\omega \\
&= e(L_\xi^\omega \delta_\chi\omega - [c, \delta_\chi\omega]) + e\delta_\chi^2\omega,
\end{aligned}$$

hence obtaining

$$\begin{aligned}
eQ_0^2\omega &= eQ_0(\iota_\xi F_\omega - d_\omega c + \delta_\chi\omega) \\
&= \frac{1}{2}e\iota_\varphi F_\omega - e\iota_\xi d_\omega\delta_\chi\omega - e[\delta_\chi\omega, c] + ed_\omega\iota_\xi\delta_\chi\omega + eL_\xi^\omega\delta_\chi\omega \\
&\quad - e[c, \delta_\chi\omega] + e\delta_\chi^2\omega \\
&= \frac{1}{2}\iota_\varphi(\text{EoM}_e) - \frac{1}{2 \cdot 3!}\bar{\psi}\iota_\varphi(\gamma^3 d_\omega\psi) \\
&\quad - \frac{1}{3!}\bar{\psi}\gamma^3\chi\left(\bar{\chi}\kappa(<\bar{e}, \underline{\gamma}d_\omega\psi>) + \frac{1}{8}\bar{\chi}\iota_{\hat{\gamma}}\iota_{\hat{\gamma}}(\underline{\gamma}d_\omega\psi)\right)
\end{aligned}$$

For c , χ and ξ , we can do the computations of Q_0^2 right away, obtaining

$$\begin{aligned}
Q_0^2 c &= Q_{PC}(\iota_\xi\delta_\chi\omega) + \delta_\chi\left(\frac{1}{2}\iota_\xi\iota_\xi F_\omega - \frac{1}{2}[c, c] + \iota_\xi\delta_\chi\omega\right) \\
&= \frac{1}{2}\iota_{[\xi, \xi]}\delta_\chi\omega + \iota_\xi L_\xi^\omega\delta_\chi\omega - \iota_\xi[c, \delta_\chi\omega] + \frac{1}{2}\iota_\xi\iota_\varphi F_\omega - \frac{1}{2}\iota_\xi\iota_\xi d_\omega\delta_\chi\omega \\
&\quad - [\iota_\xi\delta_\chi\omega, c] + \frac{1}{2}\iota_\varphi\delta_\chi\omega + \iota_\xi\delta_\chi^2\omega \\
&= \frac{1}{2}\iota_\varphi\delta_\chi\omega + \iota_\xi Q_0^2\omega,
\end{aligned}$$

having used the fact that $\frac{1}{2}\iota_{[\xi, \xi]}\delta_\chi\omega + \iota_\xi L_\xi^\omega\delta_\chi\omega - \frac{1}{2}\iota_\xi\iota_\xi d_\omega\delta_\chi\omega = 0$.

For $Q_0^2\xi$, we see

$$\begin{aligned}
Q_0^2(\xi) &= \frac{1}{2}Q_0([\xi, \xi] + \varphi) = \frac{1}{2}[\xi, \varphi] + \frac{1}{2}Q_0(\bar{\chi}\gamma^a\chi e_a^\mu)\partial_\mu \\
&= \frac{1}{2}[\xi, \varphi] + L_\xi^\omega(\bar{\chi})\gamma^\mu\chi\partial_\mu - [c, \bar{\chi}]\gamma^\mu\chi\partial_\mu - \frac{1}{2}\iota_\varphi\bar{\psi}\gamma^\mu\chi\partial_\mu + \frac{1}{2}\bar{\chi}\gamma^a\chi Q_0(e_a^\mu)\partial_\mu.
\end{aligned}$$

Now, since $e_a^\mu e_\mu^b = \delta_a^b$, we have $Q_0(e_a^\mu) = -e_a^\nu e_\nu^\mu Q_0(e_\nu^b)$, obtaining

$$\begin{aligned}
Q_0^2(\xi) &= \frac{1}{2}[\varphi, \xi] + L_\xi^\omega(\bar{\chi})\gamma^\mu\chi\partial_\mu - [c, \bar{\chi}]\gamma^\mu\chi\partial_\mu - \frac{1}{2}\iota_\varphi\bar{\psi}\gamma^\mu\chi\partial_\mu \\
&\quad - \frac{1}{2}e_b^\mu\bar{\chi}\gamma^\nu\chi((L_\xi^\omega e)_\nu^b - [c, e_\nu]^b + \bar{\chi}\gamma^b\psi_\nu)\partial_\mu \\
&= -\frac{1}{2}[\xi, \varphi] + L_\xi^\omega(\bar{\chi})\gamma^\mu\chi\partial_\mu - [c, \bar{\chi}]\gamma^\mu\chi\partial_\mu - \frac{1}{2}\iota_\varphi\bar{\psi}\gamma^\mu\chi\partial_\mu \\
&\quad + \frac{1}{2}\bar{\chi}[c, \gamma]^\mu_V\chi\partial_\mu - \frac{1}{2}\bar{\chi}\gamma^\mu\iota_\varphi\psi\partial_\mu + \frac{1}{2}\bar{\chi}L_\xi^\omega(\gamma^\mu\partial_\mu)\chi \\
&= -\frac{1}{2}L_\xi^\omega(\phi) + \frac{1}{2}L_\xi^\omega(\bar{\chi}\gamma^\mu\chi\partial_\mu) = 0.
\end{aligned}$$